

Semantic analysis

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Theory of Compilation
Laboratory 4

Type system

- safe or unsafe
- static or dynamic
- explicit or implicit

Type checking

Type checking

```
from collections import defaultdict

...

ttype['+']['int']['int'] = 'int'
ttype['+']['int']['float'] = 'float'
ttype['+']['float']['int'] = 'float'
ttype['+']['float']['float'] = 'float'
```

Type checker

Type checker

```
class TypeChecker(object):

    def visit(self, node):
        method = 'visit_' + node.__class__.__name__
        visitor = getattr(self, method,
                           self.generic_visit)

        return visitor(node)

    def visit_BinExpr(self, node):
        type1 = self.visit(node.left)
        type2 = self.visit(node.right)
        op     = node.op;
        # ...
        #
```

```
class SymbolTable(object):

    # parent scope and symbol table name
    def __init__(self, parent, name):
        ...

    # put variable symbol or fundef under <name>
    entry
    def put(self, name, symbol):
        ...

    # get variable symbol or fundef from <name> entry
    def get(self, name):
        ...

    def getParentScope(self):
        ...

    def pushScope(self, name):
        ...

    def popScope(self):
        ...
```

Break and continue

```
continue;  
break;
```

- ① <https://sly.readthedocs.io/en/latest/sly.html>
- ② <http://www.dabeaz.com/ply/ply.html>